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Investigation on the Influencing Factors of Enhanced Oil Recovery and CO_2 Storage by Impure CO_2 Flooding in Tight Oil Reservoirs

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Abstract

Impure CO_2 flooding has obvious effects on the development of unconventional reservoirs in terms of enhanced oil recovery (EOR) and CO_2 storage. Compared with impure CO_2 flooding, impure CO_2 flooding requires less CO_2 content, leading to gradually become one of the mainstream development methods for unconventional reservoirs. Nevertheless, the influencing factors of impure CO_2 flooding in tight reservoirs have rarely been studied by scholars. In this paper, by conducting impure CO_2 flooding experiments and reservoir-scale numerical simulation studies, different associated gas components, CO_2 concentrations, gas injection volumes and bottomhole pressures of production wells are investigated for the first time in impure CO_2 flooding, and the influencing laws of each influencing factor of impure CO_2 flooding on the EOR and CO_2 storage effects are elucidated. In addition, we quantify and analyze the correlation coefficients of each influencing parameter on EOR and CO_2 storage based on experiments and numerical simulations data using machine learning. The results of the core displacement experiments indicate that associated gas with high content of intermediate components promotes the EOR and CO_2 storage efficiency, enhancing the oil recovery and CO_2 storage efficiency to 83.48% and 71.31%, respectively. However, the associated gas with high content of C_1 component inhibits the EOR and CO_2 storage. Numerical simulation results of the reservoir reveal that the smaller the CO_2 concentration of impure CO_2 mixed with associated gas of high content of intermediate components, the better the oil recovery and CO_2 storage effect. The increase of impure CO_2 gas injection volume and bottomhole pressure of production wells promote the EOR effect, but are unfavorable to CO_2 storage. The machine learning results show that the gas injection volume of impure CO_2 has the greatest effect on EOR and CO_2 storage, followed by the bottomhole pressure of production wells, and the CO_2 concentration has the least effect. This paper conducts the investigation of impure CO_2 flooding in tight reservoirs from a new perspective, which has certain theoretical guidance and reference significance for impure CO_2 flooding in tight reservoirs.

Introduction

As the proven reserves of unconventional reservoirs around the world have been continuously increased, the exploitation of unconventional reservoirs has gradually become a major topic of common research by scholars around the world (Tang et al., 2022, Qi et al., 2021, Zhao et al., 2025, Davoodi et al., 2025). Unconventional reservoirs are characterized by low reservoir permeability, insufficient natural energy, and difficult to produce crude oil from the formation (Li and Cai, 2023, Han et al., 2025, He et al., 2025). It is difficult to accomplish the efficient production of unconventional reservoirs by using traditional water flooding (Xue et al., 2025, Zhang et al., 2025, Ren et al., 2025). Consequently, CO_2 flooding replaces the traditional water flooding with its high oil recovery and easy injection (Wang et al., 2025, Zhu et al., 2024, Liu et al., 2025). In addition, CO_2 flooding can realize the storage of CO_2 while displacing oil, meeting the goal of CCUS (CO_2 capture, utilization and storage) (Jia et al., 2023, Rui et al., 2025, Hao et al., 2022, Kumar et al., 2022).

Nevertheless, the transportation cost of CO_2 is high for some unconventional reservoirs that lack nearby CO_2 gas sources (Yang et al., 2024b, Chen et al., 2024). Moreover, when the formation pressure is low, CO_2 flooding is difficult to reach the miscible state, and the oil displacement efficiency is low (Chen et al., 2020, Mohammadian et al., 2024, Huang et al., 2023). As a result, in order to solve the above problems, impure CO_2 containing with associated gas has been more and more utilized in the production of unconventional reservoirs (Yu et al., 2024, Lin et al., 2024, Wu et al., 2024). Since impure CO_2 contains a considerable portion of associated gas produced in the oilfield, the oilfield where impure CO_2 flooding is applied has lower requirements for CO_2 gas source (Wen et al., 2024, Wang et al., 2024, Wu et al., 2024). When impure CO_2 contains a higher content of intermediate components, the intermediate components will reduce the minimum mixing pressure (MMP) of the injected gas-crude oil system, enhance the miscibility capacity, and further extend the time of miscible flooding (Liu et al., 2022, Yang et al., 2024a, Bon et al., 2005, Kong et al., 2021).

When intermediate components (C_2-C_6) are added to CO_2, the MMP will be reduced and oil recovery will be increased; while when light components such as C_1 and N_2 are added to CO_2, the MMP will be increased, and oil recovery will be reduced (Akpobi and Oboh, 2022, Hartman and Shu, 1987, Choubineh et al., 2019, Yu et al., 2020). Yin et al., (2014) carried out slim tube experiments and found that the smaller the concentration of CO_2 in the re-injected gas is, the higher the MMP is of the injected gas-crude oil system, and the more difficult it is to reach the mixed phase. This is since the associated gas contains a large amount of C_1, which plays an inhibitory role in the mixing of the injected gas-crude oil system, reducing the efficiency of oil displacement. In addition to C_1, Belhaj et al., (2013) found that the addition of N_2 to CO_2 increased the MMP and the addition of C_2 decreased the MMP through reservoir-scale numerical simulations. Similarly, Zhang et al., (2004) found that both C_1 and N_2 increased the MMP of the injected gas-crude oil, and the effect of N_2 was greater. Song et al., (2025) conducted two microfluidic displacement experiments of impure CO_2 containing different types of associated gas and obtained similar conclusions. They found that the addition of associated gas with higher content of intermediate components to CO_2 would promote EOR, while the addition of associated gas with higher content of C_1 would inhibit EOR. Luo et al., (2012) found through PVT experiments that the addition of a certain content of C_3 to CO_2 would enhance the solubility of the gas in crude oil, promote the viscosity-reducing effect of crude oil, increasing the oil displacement efficiency of impure CO_2.

A large number of literatures have proved that impure CO_2 can effectively reduce MMP and improve oil displacement efficiency, which is highly favorable for the development of unconventional reservoirs. However, the influencing factors of the development effect of impure CO_2 flooding are relatively vague. In this paper, we carry out the core displacement experiments and reservoir-scale numerical simulations for the first time to analyze the effects of associated gas components, CO_2 concentration, injection volume, and bottomhole pressure of production wells on EOR and CO_2 storage in impure CO_2 flooding. In addition, the weights of each influence factor on oil recovery and CO_2 storage efficiency are quantified and compared through machine learning. This paper provides some reference significance and theoretical value for the study of impure CO_2 flooding in EOR and CO_2 storage effect.

Materials and Methodology

Materials

Reservoir oil. The oil used in this experiment was derived from the crude oil of Daqing oilfield. The compounding of formation live oil was completed by compounding the produced degassed oil with associated gas according to the gas-oil ratio in the PVT report. The compounded formation oil is shown in Fig. 1. The formation live oil of this block contains a large number of intermediate and light components and belongs to typical light oil.

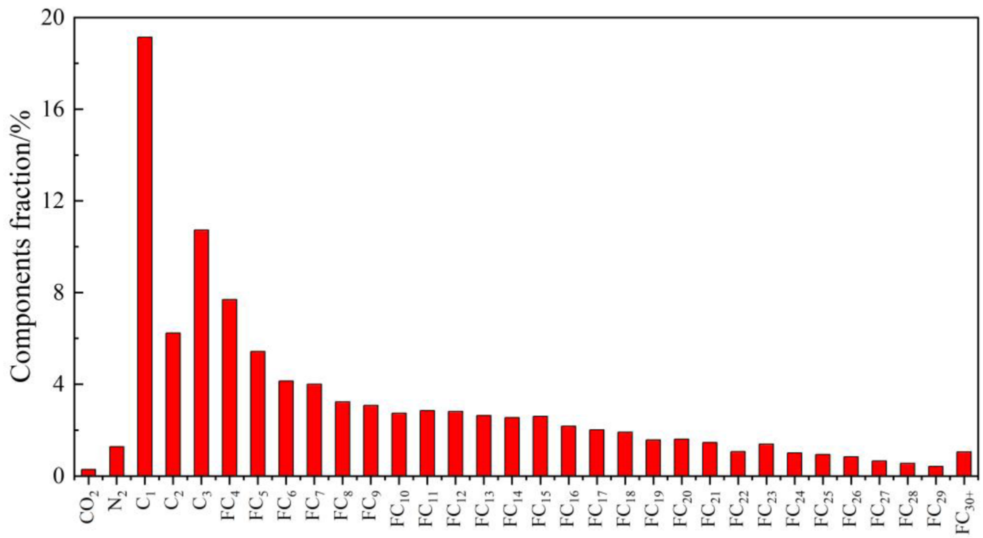


Figure 1—Composition of crude oil after compounding.

Injected gas. The gas used in this experiment is associated gas from Block X and Block Y of Daqing Oilfield. In order to better reflect the influence of different associated gas components in impure CO_2 on the development effect, we especially chose associated gas A with higher content of intermediate components and associated gas B with higher content of C_1 components. We mixed 50% CO_2 with 50% associated gas A as Gas #2, and 50% CO_2 with 50% associated gas B as Gas #3. The injected gases in this experiment are Gas #1, Gas #2 and Gas #3, and the composition of the injected gases is shown in Table 1.

Table 1—Compositions of three injected gas.

Gas	Composition/%									
	CO_2	N_2	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
#1	100	/	/	/	/	/	/	/	/	/
#2	50.286	1.517	22.285	6.85	10.201	5.416	2.051	0.832	0.384	0.177
#3	50.568	0.823	36.432	5.114	3.501	2.31	0.929	0.195	0.129	/

Core. The cores used in this experiment are also from Daqing oilfield. The core parameters are shown in Table 2. The porosity and permeability of the cores used in this experiment are low, which are typical tight cores.

Table 2—Parameters of the cores used in the experiment.

Core	Length/cm	Diameter/cm	Permeability/10^{-3}μm^2	Porosity/%
C-1	2.531	5.029	0.29	9.33
C-2	2.539	5.091	0.27	9.04
C-3	2.535	5.045	0.28	9.11

Core displacement experiments

In this study, the three gases are injected into the core saturated with oil. By calculating the oil recovery and CO_2 storage efficiency under different gases, the influence laws of different component types of associated gases in impure CO_2 on the effect of impure CO_2 flooding in EOR and CO_2 storage are clarified.

The device figure of this core displacement experiment is shown in Fig. 2. The experimental setup mainly consists of displacement equipment and a core gripper with a pressure resistance of 50 MPa and a temperature resistance of 150 °C. The displacement equipment consists of an injection pump (Teledyne Isco 500D), a circumferential pump, a pressure back valve, two intermediate vessels, a pressure monitoring system, a measuring cylinder and a gas flow meter.

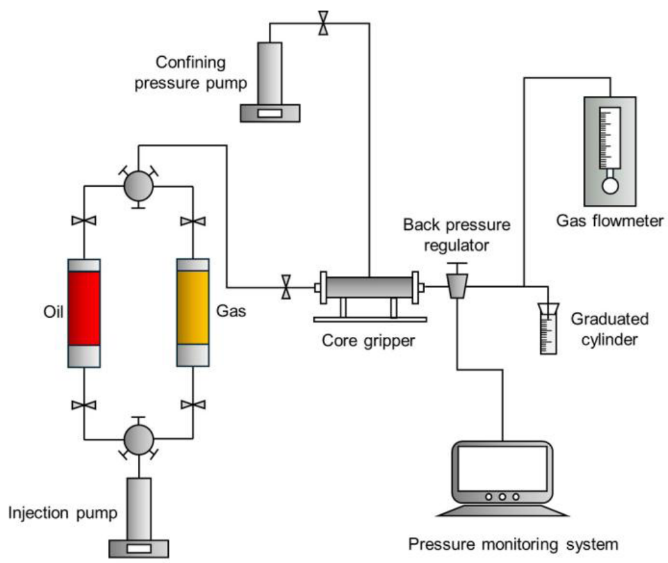


Figure 2—Schematic diagram of the core displacement experiment.

The experimental steps of the core displacement experiments are shown in Fig. 3. In order to ensure that the EOR and CO_2 storage law of impure CO_2 in the actual formation can be fully clarified, the temperature and pressure of this core displacement experiments are consistent with the temperature and pressure of the actual formation: the experimental temperature is 70 °C, and the experimental pressure is 25 MPa. The calculation of oil recovery and CO_2 storage efficiency is accomplished through the measurement of the crude oil displaced in the cylinder and the CO_2 in the gas flowmeter.

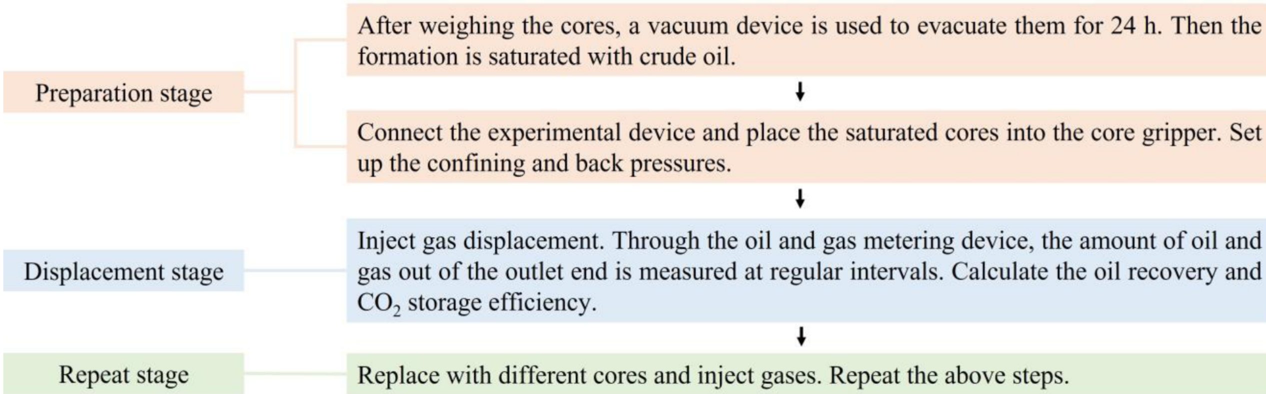
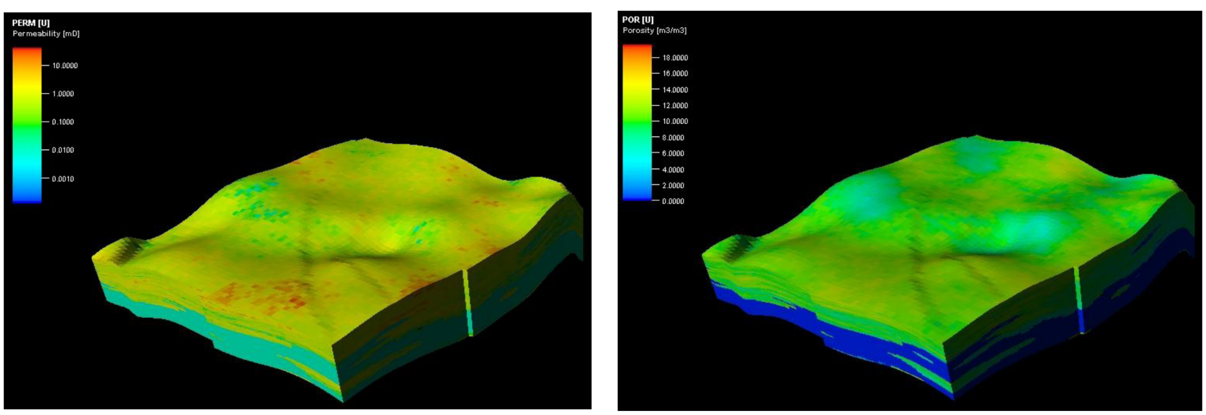


Figure 3—The process figure of core displacement experiments.

Reservoir-scale numerical simulations

In this study, a reservoir-scale numerical simulation model of the actual reservoir in Daqing oilfield is established. By comparing the recovery and CO_2 storage efficiency after 20 years development of gas injection with different CO_2 concentrations, injection volumes, and bottomhole pressures of production wells, the influence parameters of impure CO_2 on EOR and CO_2 storage efficiency are clarified.

The permeability and porosity distributions of the reservoir-scale numerical simulation model are shown in Fig. 4. To accurately represent the development effect of impure CO_2 in the actual formation, we used the component model of double-porosity and double-permeability for this numerical simulation model. The average permeability of this model is $0.28 \times 10^{-3} \mu m^2$ and the porosity is 9.1%, which is a typical model for tight reservoirs. The detailed parameters of the reservoir-scale numerical simulation model are shown in Table 3.



(a) (b)

Figure 4—Permeability and porosity distributions of the reservoir-scale numerical simulation model. (a) Permeability; (b) Porosity

Table 3—The detailed parameters of the reservoir-scale numerical simulation model.

Model Parameters	Value
Pressure/MPa	25
Temperature/°C	70.48
Porosity/%	9.1
Permeability/$10^{-3}\mu m^2$	0.28
Oil Saturation/%	66

Results and discussion

Associated gas compositions

The results of the oil recovery and CO_2 storage efficiency of the core displacement experiments of impure CO_2 under the conditions of different associated gas compositions are shown in Fig. 5 and Fig. 6. Gas#2 has the highest recovery, 83.48%; Gas#1 has the second highest recovery; and Gas#3 has the lowest recovery, 76.25%. The results of the core displacement experiments proved that the addition of associated gas with high content of intermediate components to pure CO_2 would improve the oil displacement effect and further enhance the EOR effect. Nevertheless, the addition of associated gas with high content of C_1 component to pure CO_2 is not favorable for gas displacement. In the early displacement stage, Gas#2, because of the high content of intermediate components, a large amount of injected gas is dissolved in the crude oil, and less crude oil is displaced out of the core, so the oil recovery is low compared with the other two gases. In the late stage of displacement, Gas#2 is dissolved in the crude oil with higher content, which has obvious oil viscosity-reducing effect and leads to easier displacement of crude oil from the core. Therefore, the recovery of Gas#2 rises faster in the middle and late stages of the displacement, and the final oil recovery is the highest. In terms of CO_2 storage efficiency, Gas#2 has the highest CO_2 storage rate of 71.31%. This is because substantial intermediate components make Gas#2 dissolve well, and a large amount of injected gas dissolves in crude oil. In addition, Gas#2 occupied the pore space of the extracted oil, so that a large amount of injected gas is buried into the formation, enhancing the CO_2 storage effect.

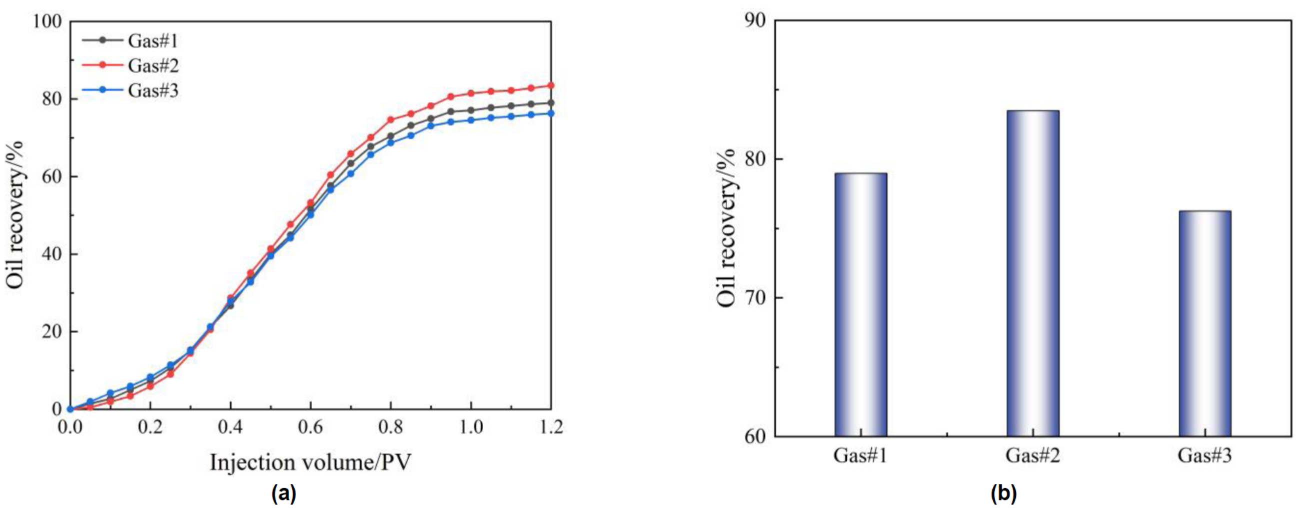


Figure 5—Oil recovery of the core displacement experiments of impure CO_2 under the conditions of different associated gas compositions. (a) Curve figure of injection volume and oil recovery; (b) Histogram of oil recovery.

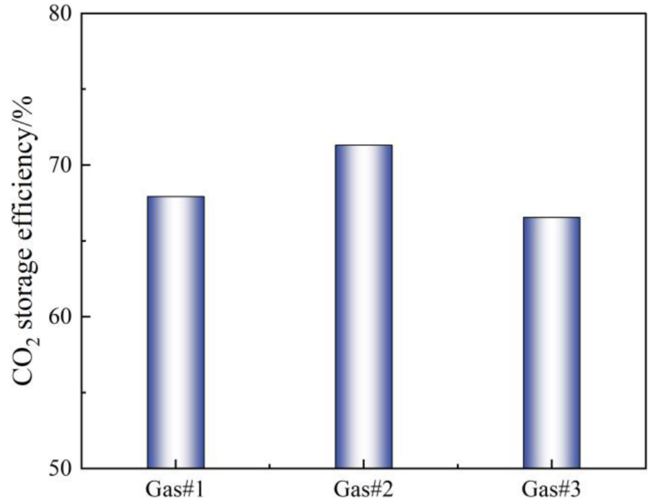


Figure 6—CO_2 storage efficiency of the core displacement experiments of impure CO_2 under the conditions of different associated gas compositions.

CO_2 concentration

In this study, Gas#2, with the best EOR and CO_2 storage effect, is selected to analyze the influence of CO_2 concentration of impure CO_2. The results of oil recovery and CO_2 storage efficiency of reservoir-scale numerical simulation with impure CO_2 under different CO_2 concentration conditions are shown in Fig. 7 and Fig. 8. After 20 years of gas injection and development, the higher the CO_2 concentration, the lower the oil recovery and CO_2 storage efficiency because of the higher content of intermediate components in Gas#2. However, there is an inflection point in the dot-line plot of CO_2 concentration versus oil recovery: when the CO_2 concentration is lower than 40%, the increase in oil recovery is lower. This is because the oil viscosity-reducing effect of Gas#2 tends to saturate when the CO_2 concentration level is too low. Therefore, continuing to add associated gas will not significantly increase oil recovery.

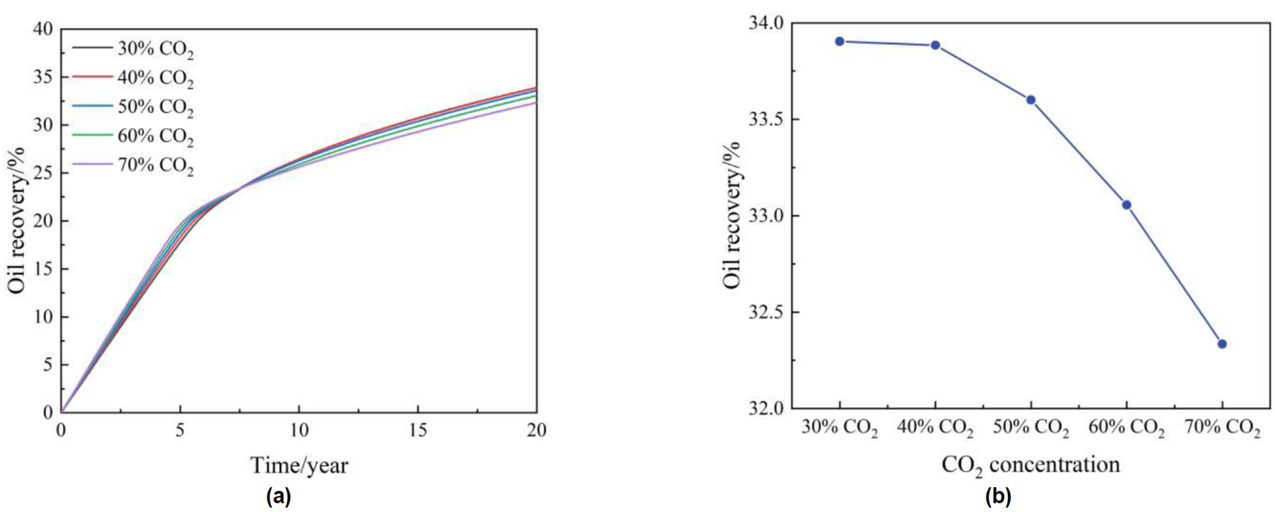


Figure 7—Oil recovery of the reservoir-scale numerical simulation with impure CO_2 under different CO_2 concentration conditions. (a) Curve figure of injection time and oil recovery; (b) Line figure of CO_2 concentration and oil recovery.

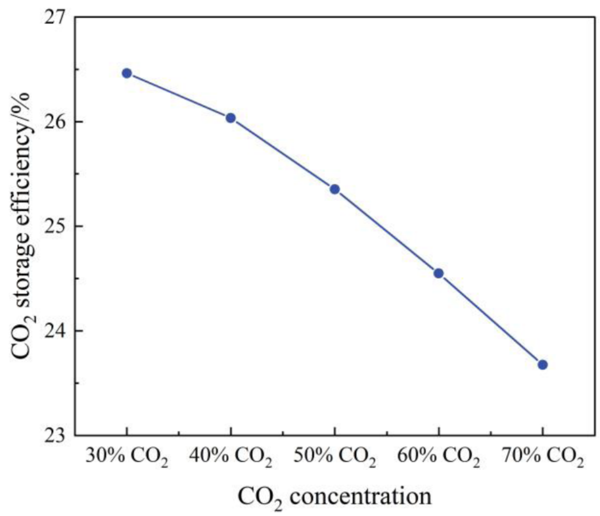


Figure 8—CO_2 storage efficiency of the reservoir-scale numerical simulation with impure CO_2 under different CO_2 concentration conditions.

Gas injection volume

In this study, Gas#2, with best EOR and CO_2 storage, is also selected to analyze the impact of gas injection volume of impure CO_2. The results of oil recovery and CO_2 storage efficiency of the reservoir-scale numerical simulation of impure CO_2 under different gas injection volume conditions are shown in Fig. 9 and Fig. 10. It is obvious that after 20 years of gas injection development, the larger the gas injection volume, the more adequate the formation energy replenishment and the higher the oil recovery. Nevertheless, a larger gas injection volume will make gas breakthrough occur earlier and be more affected by gas breakthrough. A large amount of injected gas will follow the gas flurry channel to the bottom of the production well and eventually be produced by the production well, resulting in a lower CO_2 storage efficiency.

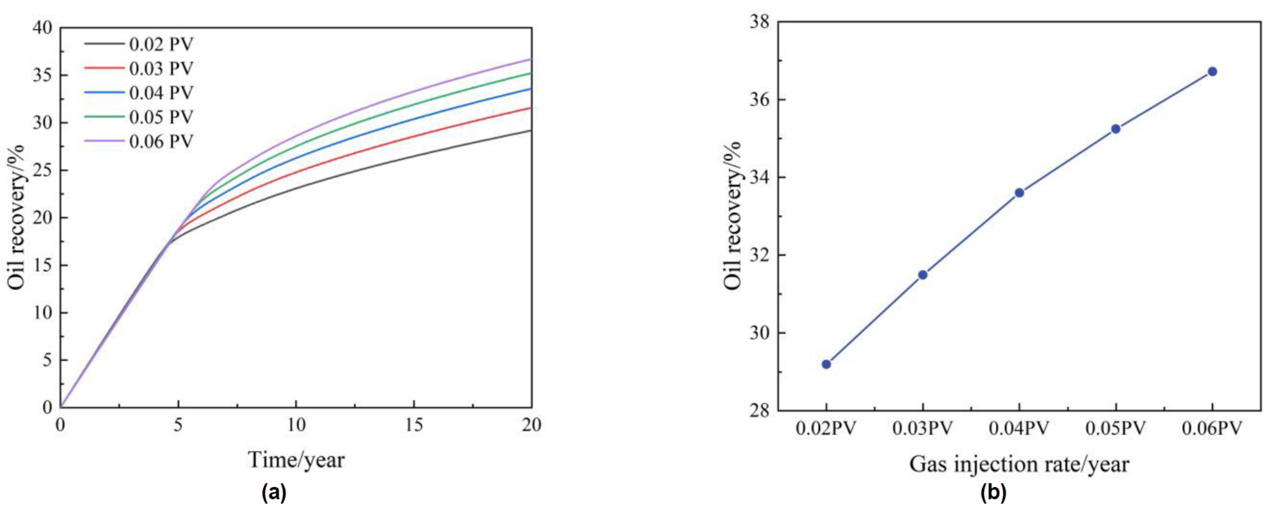


Figure 9—Oil recovery of the reservoir-scale numerical simulation with impure CO_2 under different gas injection volume conditions. (a) Curve figure of injection time and oil recovery; (b) Line figure of gas injection and oil recovery.

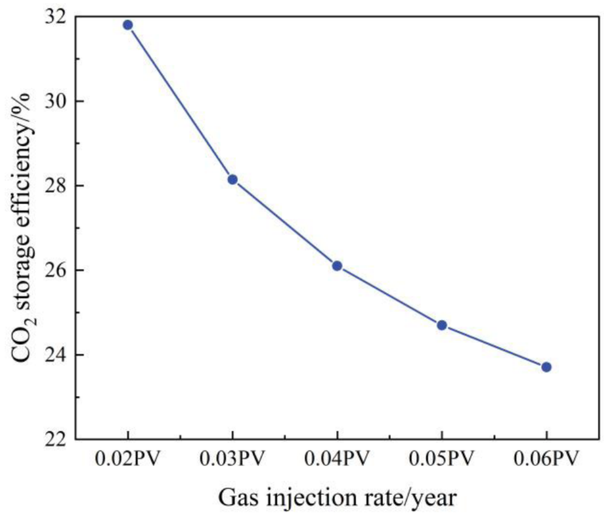


Figure 10—CO_2 storage efficiency of the reservoir-scale numerical simulation with impure CO_2 under different gas injection volume conditions.

Bottomhole pressure of production wells

In this study, Gas#2, with best EOR and CO_2 storage effect, is still chosen as well to analyze the effect of bottomhole pressure of production wells with impure CO_2. The oil recovery results of reservoir-scale numerical simulations of impure CO_2 under different bottomhole pressure of production wells conditions are shown in Fig. 11 and Fig. 12. When the bottomhole pressure of the production well is low, the production differential pressure of the gas injection development process increases, raising the final oil recovery. However, the lower bottomhole flow pressure in the production well enhances the flow of formation fluids to the production well. More of the injected gas is transported to the bottom of the production well and is produced, resulting in lower CO_2 storage efficiency.

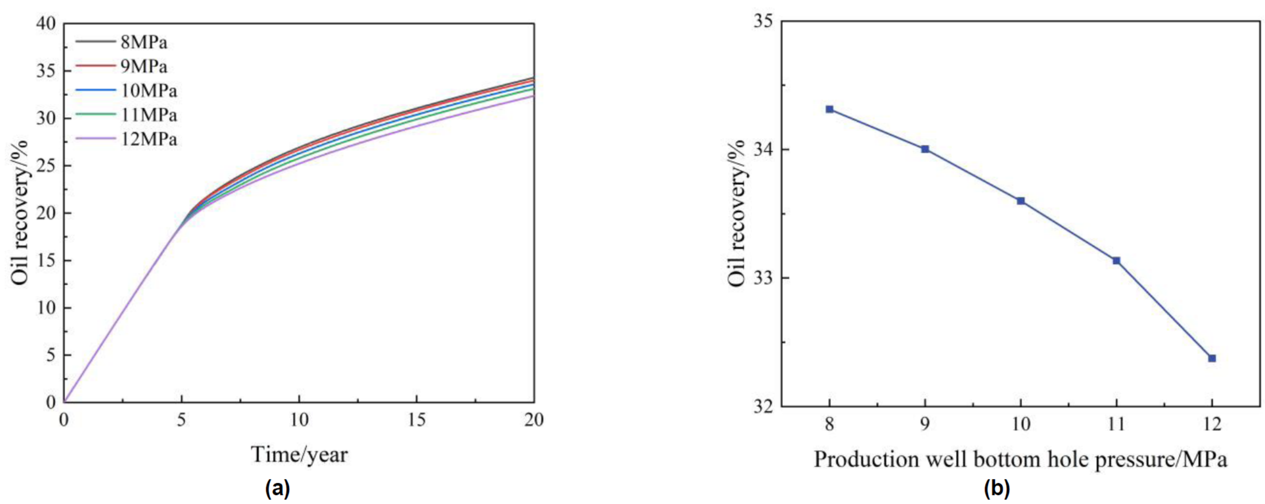


Figure 11—Oil recovery of the reservoir-scale numerical simulation with impure CO_2 under different bottomhole pressure of production wells conditions. (a) Curve figure of injection time and oil recovery; (b) Line figure of production wells bottomhole pressure and oil recovery.

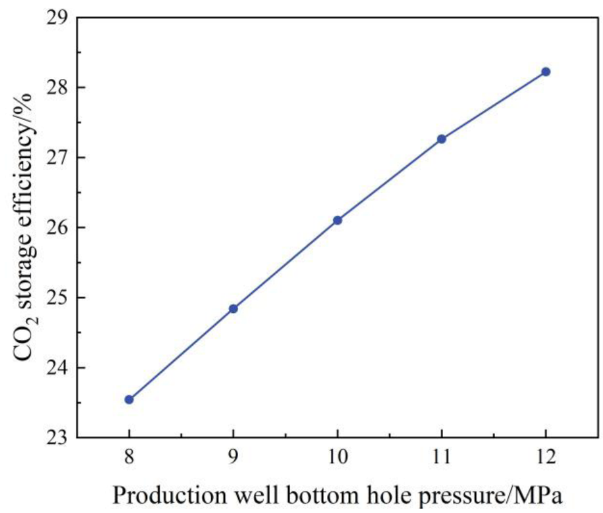


Figure 12—CO_2 storage efficiency of the reservoir-scale numerical simulation with impure CO_2 under different bottomhole pressure of production wells conditions.

Analysis of influencing factors

In this study, a quantitative analysis of the weights of the individual influencing factors of impure CO_2 on EOR and CO_2 storage effect is completed by using Pearson correlation analysis. The heat map of each influencing factor of impure CO_2 on EOR and CO_2 storage effect is shown in Fig. 13. The most influential factor on EOR and CO_2 storage efficiency is the gas injection volume, with correlation coefficients of 0.782 and 0.667. This is because the gas injection volume determines the amount of supplemental formation energy, whether it can improve the effect of gas injection and the size of the impact of gas breakthrough. The bottomhole pressure of the production wells has the second highest degree of influence on the EOR and CO_2 storage effect. CO_2 concentration has the least influence on EOR and CO_2 storage effect, with correlation coefficients of 0.396 and 0.368. This study proves that the CO_2 concentration of impure CO_2 can be adjusted to a certain extent in order to save the cost of injecting gas into the oilfield.

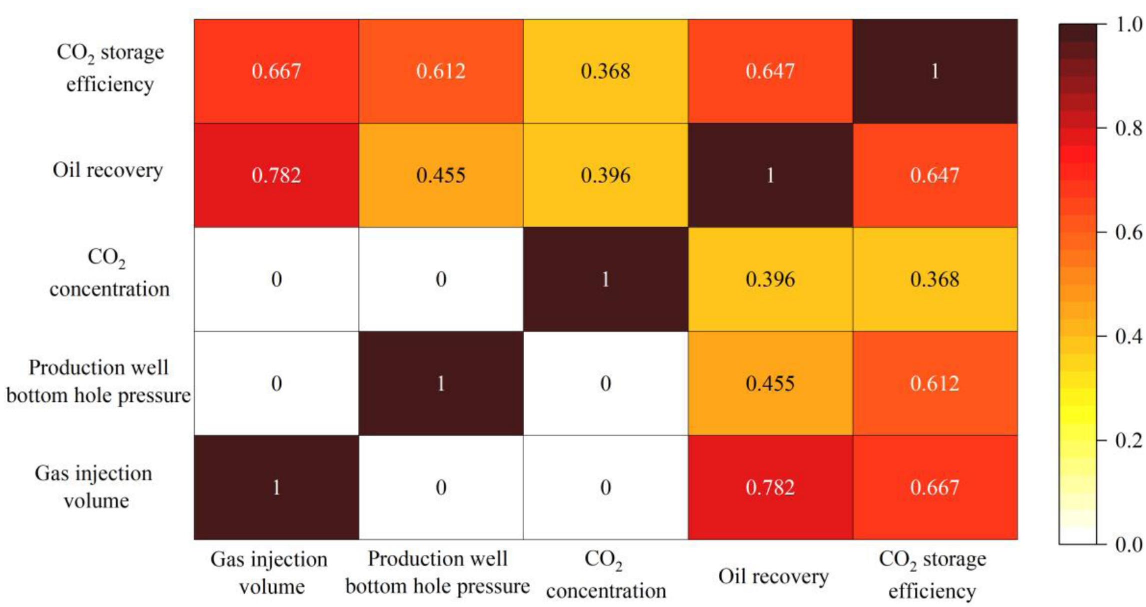


Figure 13—Heat map of each influencing factor of impure CO_2 on EOR and CO_2 storage effect.

Conclusions

In this paper, the influencing factors of impure CO_2 flooding in tight reservoirs are investigated for the first time by conducting impure CO_2 core displacement experiments and reservoir-scale numerical simulations. The influencing laws of gas components, CO_2 concentration, gas injection volume and bottomhole pressure of production wells on impure CO_2 flooding in EOR and CO_2 storage are clarified. In addition, each influencing factor is quantified and analyzed through carrying out machine learning. This study has certain theoretical value and reference significance for the investigation of impure CO_2 flooding in tight reservoirs. The conclusions of this paper are as follows:

1. The associated gas with higher content of intermediate components in impure CO_2 promotes the diffusion of injected gas into crude oil, which facilitates the EOR and CO_2 storage effect, and enhances the oil recovery and CO_2 storage efficiency to 83.48% and 71.31%. However, the associated gas with high content of C_1 component in impure CO_2 inhibits the EOR and CO_2 storage effect, resulting in only 76.25% and 66.54% oil recovery and CO_2 storage efficiency.
2. Under different CO_2 concentration conditions in impure CO_2, the lower the CO_2 concentration, the better the EOR and CO_2 storage effect. This is caused by the associated gas in impure CO_2 containing higher intermediate components. Under different gas injection and production well pressure conditions for impure CO_2, higher gas injection and lower production well pressure both lead to higher oil recovery, but inhibit the CO_2 storage effect.
3. Among the influencing factors of impure CO_2, the gas injection volume has the greatest effect on EOR and CO_2 storage, with Pearson's coefficients of 0.782 and 0.667, respectively. The bottomhole flow pressure of the production wells is the second highest. While the CO_2 concentration has the smallest effect on EOR and CO_2 storage.

Acknowledgements

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